

Gold Mining in the United States – Technical Report

Executive Summary: The United States produces on the order of 6–7 million troy ounces (200–220 tonnes) of gold per year, with cumulative historical output exceeding 10,000 tonnes. Production peaked during the California Gold Rush (1849–50) and again in the 1980s with the advent of large open-pit heap-leach mining in Nevada. In 2019 the US produced about 6.4 million ounces (200 t) of gold

【84†L210-L213】 (from 12 states) – making it the fourth-largest gold producer globally – with Nevada alone accounting for about 72% of U.S. output 【7†L50-L56】 . Major producing states today are Nevada (by far the largest), Alaska (~8–9% of U.S. gold in 2019 【84†L210-L213】), and Colorado (third-ranked state, ~360,000 oz in 2018 【87†L258-L266】). The richest current mines are large Nevada open pits (Carlin-Trend and epithermal deposits) and Alaskan high-grade underground and open-pit mines (e.g. Fort Knox, Pogo, Kensington). Proven and probable (“reserves”) plus measured and indicated resources exceed tens of millions of ounces in several districts: for example, the Donlin Creek (Alaska) deposit alone hosts ~33.8 Moz proven+probable within a 39 Moz measured+indicated resource 【84†L293-L298】 . Proposed and undeveloped projects like Donlin Creek (Alaska) and others in Nevada (e.g. the Hycroft, Sleeper and Sandman deposits) add dozens more Moz of gold endowment 【59†L179-L186】 【84†L293-L298】 .

Regional geology varies: **Nevada’s** gold is largely sediment-hosted (Carlin-type and related epithermal) in Paleozoic carbonate rocks, controlled by faulting and hydrothermal alteration. **Alaska’s** production is from high-grade mesothermal vein deposits (e.g. Pogo, Kensington), porphyry-related systems (Donlin), and placer deposits (historic Nome, Fairbanks gold placers). **Colorado’s** gold comes from alkaline igneous complexes (Cripple Creek & Victor, low-sulfidation epithermal in older diatremes). Other deposit types include orogenic (lode gold in metamorphic belts, e.g. historical Carolinas production), porphyry-associated gold (as Cu-Au byproduct, e.g. Arizona copper mines), skarn and polymetallic (e.g. Greens Creek, AK). Geologically, >75% of U.S. lode-gold is in hydrothermal deposits (Carlin-type, epithermal, orogenic) that form in orogenic and extensional belts. Exploration relies on geologic mapping, detailed structural analysis, and geochemical/alteration footprints (e.g. As-Sb-Hg pathfinders in Carlin systems). Geophysical methods (e.g. gravity, induced polarization, magnetics) help delineate blind deposits.

Environmental and permitting constraints are significant. Most of Nevada’s gold is on federal land (BLM/USFS), requiring multi-agency NEPA reviews (EIS) and state permits. Gold mining uses cyanide (heap-leach) or pressure oxidation, so Clean Water Act (NPDES for tailings, 303d for water quality) and Clean Air Act (emissions) apply. States (e.g. NV, AK, CO) also enforce reclamation bonds and water rights. Social and cultural issues (e.g. subsistence in Alaska, tribal lands) can affect projects. By contrast, much of the 19th-century gold era (e.g. California placers) was on private land with fewer regs.

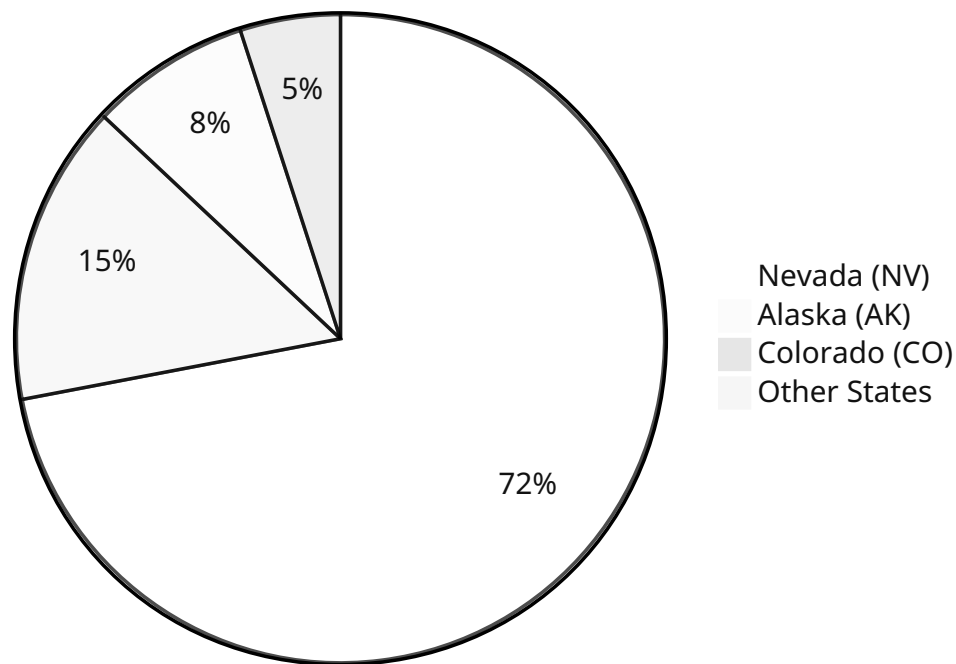
Infrastructure in leading districts is generally good: paved roads, power, skilled labor and equipment suppliers cluster around Nevada’s gold camps and Alaska’s Interior. Remote Alaskan projects rely on local airstrips/ports and seasonal road access. Water supply and power can be limiting factors (e.g. arid Nevada). Proximity to processing centers (for concentrates or cyanide circuits) is important.

Economically, large modern gold mines are low-grade but high-tonnage. Nevada open-pit mines often operate at grades of ~0.01–0.03 oz/ton (0.3–1 g/t) with heap-leach recovery (~85–95%). Colorado’s

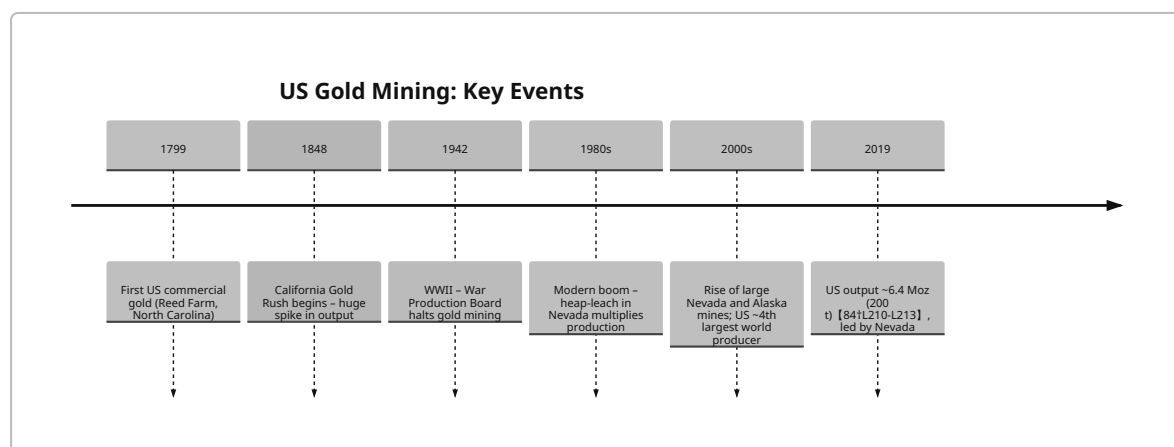
Cripple Creek (alkaline O/C) processes ore ~0.02 oz/ton (0.67 g/t) by flotation and roasters 【87†L288-L295】. Alaska's underground mines average much higher grades (e.g. Kensington ~8 g/t in mineable zones) but with higher operating costs. Typical all-in-sustaining costs (AISC) for U.S. gold are roughly \$1,000–1,500 per ounce (depending on scale and grade). Capital costs for a new mid-size open-pit project range from hundreds of millions to >\$1 billion.

State Production Chart: The chart below shows the approximate share of U.S. gold output by state. Nevada dominates (≈70–75% 【7†L50-L56】), followed by Alaska (~8–10% 【84†L210-L213】) and Colorado (~5%). The “Other” category includes small contributions from states like Arizona, Utah, California and Idaho.

US Gold Production by State (approx. %)



Historical Production (Timeline)



Major Producing States and Districts

- **Nevada:** By far the leading producer ($\approx 72\%$ of U.S. gold [\[7†L50-L56\]](#)). Major districts include the Carlin Trend (Goldstrike, Carlin, Twin Creeks, Phoenix, etc.), Cortez-Carlin (Cortez Hills, Pipeline), Round Mountain (low-sulfidation epithermal), and various other sedimentary/epithermal systems. These are typically large open-pit, low-grade heap-leach operations owned by Barrick/Newmont (Nevada Gold Mines JV), Kinross, SSR, and others.
- **Alaska:** Second-ranked ($\approx 8\%$ of US in 2019 [\[84†L210-L213\]](#)). Top mines (Fort Knox, Pogo, Kensington, Greens Creek) are underground or open pit. Fort Knox (Kinross) is an open-pit disseminated deposit averaging $\sim 0.37\text{--}0.40$ g/t [\[84†L239-L244\]](#) ; Pogo (Northern Star) is a high-grade ($\sim 8\text{--}15$ g/t) vein gold mine; Kensington (Coeur) is a high-grade mesothermal vein deposit with 2015 reserves ≈ 0.56 Moz [\[84†L259-L264\]](#) ; Greens Creek (Hecla) is a polymetallic VMS, 2019 produced ~ 56.6 k oz Au [\[84†L269-L272\]](#) (mostly silver). Cumulative Alaska production since 1880 is ~ 49.3 Moz [\[84†L219-L222\]](#) .
- **Colorado:** Third-ranked (2018: 360,000 oz [\[87†L258-L266\]](#)). Virtually all current gold is from the Cripple Creek & Victor open-pit mine (Coeur Mining/Centerra), an alkaline igneous hosted deposit. In 2019 CC&V produced $\sim 322,000$ oz [\[87†L288-L295\]](#) from ore grading ~ 0.67 g/t Au. Historically, Colorado has many gold districts (Leadville, Central City, Silverton, etc.), but modern output is focused on CC&V.
- **Others:** Smaller producers include (2019 data) Utah (~ 40 k oz), South Dakota (homestake closed; minor output from surface), California (a few small lode/placer mines), New Mexico, etc. Historically important gold states (e.g. Virginia, Georgia, North Carolina) are now negligible producers.

Mine-Level Data (Major Mines)

Mine (District)	Operator	2019 Production	Reserves/Resources	Status
Fort Knox (Fairbanks, AK)	Kinross Gold	6,229 kg (200,263 oz) [84†L239-L244]	59,000 kg (1.9 Moz) Reserves [84†L239-L244]	Operating (open-pit)
Pogo (Goodpaster, AK)	Northern Star Resources	4,808 kg (154,589 oz) [84†L247-L252]	— (currently near mine life)	Operating (underground)
Kensington (Berners Bay, AK)	Coeur Mining	3,979 kg (127,914 oz) [84†L259-L264]	17,000 kg (0.56 Moz) Reserves [84†L259-L264]	Operating (underground)
Greens Creek (Juneau, AK)	Hecla Mining	1,761 kg (56,625 oz) [84†L269-L272]	(gold as byproduct; Ag-focused)	Operating (underground)

Mine (District)	Operator	2019 Production	Reserves/Resources	Status
CC&V (Cripple Creek, CO)	Newmont (was Coeur)	9,992 kg (322,000 oz) 【87†L288-L295】	(≈9.7 Moz Au in R+P 【87†L288-L295】)	Operating (open-pit)
Hycroft (Nevada)	SSR Mining (held by Hycroft Mining)	0 (care & maintenance)	Measured+Indicated: 16.4 Moz Au 【59†L179-L186】	Care/ Maintenance

Sources: Production figures from company reports and USGS summaries 【84†L239-L244】 【84†L247-L252】 【87†L288-L295】 . Reserves/resources from mine reports 【84†L239-L244】 【84†L259-L264】 【59†L179-L186】 .

Deposit Types and Controls

Gold in the U.S. occurs in a variety of deposit types:

- **Carlin-Type (Sediment-Hosted):** Fine-grained gold in sedimentary rocks (e.g. silty limestones) of Paleozoic age, especially in Nevada's Carlin Trend and Eureka district. Controlled by deep extension and hydrothermal alteration zones. These deposits yield bulk-tonnage, low-grade ore (~0.2–2.0 g/t Au) with arsenic-antimony telluride pathfinder minerals. Major examples: Goldstrike, Cortez, Rain mining districts.
- **Epithermal (Volcanic-Hosted):** Intermediate-sulfidation and low-sulfidation quartz-adularia vein systems in Tertiary volcanic terranes. Found in Nevada (e.g. Round Mountain, Getchell) and other mountainous regions. Generally higher grade (1–10 g/t) but smaller tonnage, controlled by caldera ring faults or extensional fractures.
- **Orogenic/Vein (Mesothermal):** Gold-quartz veins in metamorphic belts. Historically important in Appalachian (e.g. Georgia, North Carolina in 19th century) and Rocky Mountain regions. Alaska's Pogo and Kensington are examples of high-grade orogenic-type deposits (vein arrays in metamorphic or intrusive rocks).
- **Placer (Alluvial):** Gold eroded from hardrock and deposited in streams/alluvium. Rich placers occurred in California, Alaska (e.g. Nome, Fairbanks) and the Arctic. Modern placer mining is minor in the US but Alaska still has some dredging.
- **Porphyry/Massive Sulfide:** Copper porphyry systems (e.g. Arizona's Morenci, Bagdad) contain significant gold as byproduct. Large skarn and volcanogenic massive-sulfide (e.g. Hecla's Greens Creek) deposits also carry gold.

Geologic controls: Gold is usually associated with major crustal structures (faults, folds), intrusions, and fluid pathways. In Nevada, gold is linked to the Basin-and-Range extensional fault systems. In Alaska, mineral belts (like the Goodpaster-Tintina) reflect arc collision events that created shear zones and granitoid intrusions. Alteration assemblages (silicification, sericitization, decalcification in carbonates) are key exploration guides. Geochemical signatures vary: Carlin-type show As–Sb–Hg–Tl anomalies; epithermal show Ag–Cu–Pb–Zn associations; porphyry show Cu–Mo–Au pathfinders.

Exploration Methods and Signatures

Exploration combines **remote sensing and geophysics** (airborne magnetics, gravity and radiometrics to map structures and intrusions) with **ground surveys**. Key methods include:

- **Mapping and Sampling:** Geological mapping to identify alteration zones, veins and host rocks. Multi-element geochemical sampling (soils, rocks) targeting gold and indicator elements.
- **Geophysics:** Gravity and magnetic surveys to locate deep alteration or intrusions; Induced Polarization (IP) surveys to outline sulfide zones; Resistivity to detect silicification halos; radiometric surveys for potassic alteration (e.g. K-Ar elements). Deep mineralization (Carlin) may require deep crustal-scale gravity or tilt-type methods.
- **Drilling:** Confirmation by diamond or RC drilling is essential. Core studies reveal ore minerals and structure. Drilling can be expensive due to depth (Nevada Carlin targets are often >200 m deep).
- **Computer Modeling:** 3D modeling of ore bodies using geologic and drill data to estimate resources (using *Measured/Indicated/Inferred* classification) and guide step-out drilling. GIS and structural modeling are widely used.

Unique geochemical signatures guide each deposit type. For instance, Nevada Carlin-type deposits lack visible gold but have microscopic Au with pyrite; they yield strong antimony, mercury, thallium anomalies. Epithermal systems often have pathfinder Ag, copper, lead, zinc. Placer exploration relies on stream sediment panning, heavy-mineral analysis and geochemical dispersion along paleochannels.

Environmental and Permitting

Gold mining in the U.S. faces stringent environmental controls:

- **Federal Permits:** On federal lands (Nevada, Alaska, parts of West), mining plans require NEPA environmental impact analyses. The EPA (Clean Water Act) regulates tailings discharge (NPDES permits) and reclaims acid-rock drainage. BLM/USFS issues Plans of Operation and must approve reclamation bonds. Endangered Species Act and cultural/historical preservation (NHPA) also apply.
- **State/Local Regulation:** States require water rights (especially for heap leach water use), air permits for dust and emissions, and oversee reclamation. For example, Nevada enforces *NRS* Title 39A, requiring a water pollution permit before mining with cyanide. Alaska requires an AQ permit and state reclamation plans under ADNRC. Counties/municipalities sometimes impose zoning or special assessments (e.g. surcharges on tonnage for schools).
- **Liability and Reclamation:** Modern mines must post large financial assurances. Historic sites (e.g. defunct dredge fields in Alaska or mercury-laden California streams) highlight legacy cleanup issues. Regulatory risk can delay or halt projects; for instance, the proposed Pebble (Alaska) Cu-Au project faced multi-year EPA review of potential watershed impacts.

Infrastructure and Logistics

Major U.S. gold districts benefit from good infrastructure. Nevada mines have year-round access via highways, on-site power, and skilled mining contractors. Concentrate/refining facilities are within North America (e.g. Pueblo, CO; Reno, NV). Alaska's remote camps (e.g. Pogo) often use private airstrips and fly-in crews; Fort Knox and Kensington receive ore by road. Ports (Skagway for Southeast AK, or Seattle ports for interior barge routes) support Alaskan operations. Water is scarce in Nevada's desert basins; mines rely on groundwater pumping or pipelines. Long-distance shipping of concentrates (to refineries) and reagent supply (cyanide, lime) are routine but add cost and logistics complexity. In contrast, eastern U.S. gold areas often lack infrastructure and face community opposition, limiting new projects there.

Economics

Grades and costs vary widely by mine type. Open-pit heap-leach gold requires very low operating cost (~\$1–3 per tonne mined) but yields only fractions of a gram per tonne. Underground vein mines (like Pogo) have high grade but high mining and development costs (AISC >\$1,000/oz). Typical recovery is 80–90% via gravity/CIL/CIP for non-refractory ore; refractory sulfide ores (e.g. Carlin, some Alaskan deposits) may need roasting or pressure oxidation (~60–85% recovery) and thus higher costs. Recent data suggest large Nevada mines have an AISC in the range of ~\$1,100–1,300/oz, while high-grade Alaskan and Colorado mines often achieve lower AISCs (\$800–\$1,100/oz) due to higher grades.

Capital costs for expanding or new operations are very high: e.g., a new large heap-leach mine (50 ktpd) can cost ~\$800M–\$1B to build (mineral processing, tailings dam, infrastructure). Exploration costs per ounce discovered are also high: drilling to intercept deep Carlin deposits can run >\$10,000 per hole. Economic viability thus hinges on scale (low unit costs) and gold price (currently ~\$1,900/oz in 2025). By comparison, average ore grades: Nevada open pits (~0.4 g/t), CC&V (0.67 g/t) **【87†L288-L295】**, Alaskan epithermal/mesothermal (several g/t), and some lode mines exceed 10 g/t.

Proven vs Potential Reserves and Resources

The U.S. classifies gold resources as “Reserves” (proven + probable) or “Resources” (measured/indicated/inferred). Proven gold reserves in US mines are on the order of tens of millions of ounces (e.g. Nevada mines collectively have ~50 Moz proven+probable as of the latest reporting). However, much more gold is identified in resources and undiscovered potential. The table below highlights selected districts and projects (current *resources* and *reserves*, with confidence):

District/ Project	Location	Proven+Prob (Moz)	Meas.+Ind. (Moz)	Inferred/ Potential (Moz)	Confidence / Source
Donlin Creek	Alaska (ILB)	33.8 (at 2.09 g/t) 【84†L293- L298】	39.0 (at 2.03 g/t) 【84†L293- L298】	+10–20 (undiscovered along trend)	Reserves estimated (2022 NI 43-101)
Hycroft (Goldbanks/ Sandman)	Nevada (Carlin)	–	16.4 (M&I) 【59†L179- L186】	+1.0–2.0 (inferred, Sandman inf. ~0.27 Moz 【59†L191- L195】)	Company resource estimates (2025)

District/ Project	Location	Proven+Prob (Moz)	Meas.+Ind. (Moz)	Inferred/ Potential (Moz)	Confidence / Source
Sleeper (Paramount Pogo)	Nevada (Carlin)	–	1.90 (M&I) 【59†L179- L186】	small (exploration stage)	Published resource (2023)
Other Nevada targets	Carlin Trend	–	(various tens to hundreds of koz)	(prospects with uncertain tonnage)	Historic estimates; high uncertainty
Alaska (total)	Alaska	~12.0 (state total, ~2020s)	~30–40 (incl. Donlin)	Likely >50 (undiscovered greenfields)	USGS and USGS estimates

Data sources: Donlin Creek (P+P and M&I from NI 43-101 filings 【84†L293-L298】); Hycroft (combined measured+indicated for Goldbanks and Sandman units 【59†L179-L186】 【59†L191-L195】); other data from company reports and USGS. Note that “potential” is highly speculative; for example, USGS mineral resource assessments suggest large undiscovered gold endowment in Alaskan terranes (e.g. Kuskokwim belt) and in Nevada basins adjacent to known deposits.

Future Exploration Targets and Recommendations

Based on geology and resources, high-potential targets include:

- **Northern Nevada Deep Targets:** The extendable Carlin trend has untested blind continuations. Explore south of Crescent valley, in peripheral basins (e.g. Cortez trends). Gravimetric and deep IP surveys may locate new Carlin orebodies beneath cover.
- **Elko County, Nevada:** Young lode systems (Sleeper, Sandman, Hycroft) suggest more shallow epithermal/circumneutral deposits. Follow-up on geochemical anomalies east of existing mines (e.g. the Black Star project area has shown 14+ g/t Au surface samples 【59†L75-L82】). Drill programs in these greenfield zones could find feeder systems.
- **Alaska Northward:** Beyond Donlin, look along the Tintina Gold Province (Fairbanks, Goodpaster districts) for new lode systems. The newly opened Manh Choh/Peak mines (Yukon-Tanana) show that orogenic veins persist. Identify structural traps near Donlin and Crystal Creek trends for deeper drilling.
- **Nevada Carlin Extensions:** Explore under-served basins (e.g. Pinson zone west of Carlin, Jerriitt Canyon extensions). Use directional drilling from existing pads to test blind targets.
- **Other Regions:** Colorado’s unexplored high-country (e.g. craton-margin intrusions) and the historical Maine, California, or Appalachia belts (low infrastructure) may host pockets of gold – but likely uneconomical at current gold prices.

Next steps: For each high-potential target, develop a systematic exploration program: detailed surface mapping, geochemistry (including whole-rock pathfinders), geophysics, and scout drilling. Early-stage metallurgical tests should accompany drilling to assess recoveries (e.g. oxide vs refractory). Economic

viability studies (NI 43-101 style) should be initiated once a significant resource is outlined. Given regulatory complexity, engaged permitting teams early is crucial, especially for projects on federal land.

Glossary (Key Terms):

- **Ore Reserve (Proven/Probable):** The economically mineable portion of a Measured/Indicated mineral resource, under current conditions.
- **Measured/Indicated/Inferred:** Resource classification by confidence; Measured is highest confidence, Inferred lowest.
- **Carlin-Type Deposit:** Sedimentary rock-hosted disseminated Au deposit (as at Nevada's Carlin Trend). Often "invisible" gold in fine pyrite.
- **Epithermal Deposit:** Shallow (<1.5 km) hot-spring related vein system; can be high-sulfidation or low-sulfidation types.
- **Orogenic (Mesothermal) Deposit:** Gold-bearing quartz-carbonate veins formed in mountain-building (orogenic) settings, typically at mid-crust depths.
- **Heap Leach:** Processing method where crushed ore is stacked and irrigated with cyanide solution to extract gold.
- **Autoclave (Pressure Oxidation):** High-pressure vessel used to oxidize and liberate gold from refractory sulfide ores.
- **All-In Sustaining Cost (AISC):** Industry metric of full production cost per ounce (includes operating costs + sustaining CAPEX).
- **NI 43-101:** Canadian reporting standard for mineral projects (used internationally).
- **Mining Claim:** A parcel of land staked for mineral rights (U.S. BLM system).
- **NEPA:** National Environmental Policy Act, requiring impact assessments for federal actions (mining on federal land).

References: Authoritative sources including USGS and state reports. Key data on production and resources are from USGS/NMIC publications [84†L210-L213] [7†L50-L56] [87†L258-L266] and company disclosures [84†L239-L244] [59†L179-L186] . Historical context from USGS geologic reviews. (All data cited above.)
